Zazr and VAVR Tour

Season: 0 · Intro · Category: FP Libraries · Level: Beginner

Intent

- Introduce Zazr (primary) and VAVR (comparison) control types and collections.

Problem

- Without this pattern, code tends toward: New things up with conditionals and duplicated construction.

Real-world Analogy

- Two toolboxes: VAVR is the classic; Zazr is the Java 25 redesign (sealed Option/Either/Try/Validation).

When to Use

- Writing null-hostile, immutable, effect-aware Java.
- The design force named in Intent shows up more than once.

When Not to Use

- Mixing both libraries in one module without a clear boundary.
- A simpler language feature (record, sealed switch, function value) already solves it.

Pattern Structure

- See mermaid diagram in explanation.md

Classic Implementation

```java
package com.buildwithwaqar.patterns.functional.vavrcompare;

/** Side-by-side Option usage — VAVR (classic) vs Zazr (Java 25 sealed/records). */
public final class VavrVsZazr {
    private VavrVsZazr() {}

    public static String vavrDescribe(io.vavr.control.Option<Integer> opt) {
        return opt.map(n -> "got " + n).getOrElse("missing");
    }

    public static String zazrDescribe(dev.zazr.control.Option<Integer> opt) {
        return switch (opt) {
            case dev.zazr.control.Option.Some(var n) -> "got " + n;
            case dev.zazr.control.Option.None() -> "missing";
        };
    }
}
```

Modern Java 25 / Functional

- Pattern-match Zazr Option.Some / Option.None; VAVR uses map/getOrElse.

Worked Example

```
See the JUnit test beside the implementation.
Run: mvn -Dtest=*Test test
Focus package matching: functional/vavrcompare/VavrVsZazr.java
```

Pitfalls

- Zazr Option rejects null — use Unit markers, not null.
- Prefer Validation.zipWith to collect errors.

Interview Q&A

- Q: Why Zazr over VAVR on Java 25? A: Sealed records + exhaustive switch, Scala 2.13 collections, ZIO-style names, no runtime deps.
- Q: How would you test this? A: Prefer injecting interfaces or interpreters; keep demos deterministic (fixed clocks, canned inputs).

Key Takeaways

- Introduce Zazr (primary) and VAVR (comparison) control types and collections.
- Prefer Java 25 sealed types / records / virtual threads where they simplify the classic shape.
- Ship working tests with every episode.

Video Script Highlights

- [0:00] Hook — Open with the analogy: Two toolboxes: VAVR is the classic; Zazr is the Java 25 redesign (sealed Option/Either/Try/Validatio...